IDA-PBC Controller Design for a Segway-like System

Overview

In this document we present an Interconnection and Damping Assignment Passivity-Based Control (IDA-PBC) approach for a Segway-like system with the dynamics derived from the following equations. The model is given in the form

$$M(q)\,\ddot{q} + C(q,\dot{q})\,\dot{q} + G(q) = B\,F,$$

where the generalized coordinate is

$$q = \begin{bmatrix} x \\ \theta \end{bmatrix}.$$

System Dynamics

Inertia Matrix

The inertia matrix is given by

$$M(q) = \begin{bmatrix} a & b\cos\theta \\ b\cos\theta & c \end{bmatrix},$$

with the parameters defined as

$$a = 2m_w + m_p + \frac{2I_W}{r^2}, \quad b = m_p\,l, \quad c = m_p\,l^2 + I_p.$$

Coriolis and Gravitational Terms

The Coriolis–centrifugal term is

$$C(q,\dot{q})\,\dot{q} = \begin{bmatrix} -b\sin\theta\,\dot{\theta}^2 \\ 0 \end{bmatrix},$$

and the gravitational term is

$$G(q) = \begin{bmatrix} 0 \\ -m_p\,g\,l\,\sin\theta \end{bmatrix}.$$

The control input enters through

$$B = \begin{bmatrix} 1 \\ 0 \end{bmatrix},$$

so that only the horizontal motion is directly actuated.

Thus, the equations of motion become:

$$\text{Horizontal (actuated) coordinate:} \quad a\,\ddot{x} + b\cos\theta\,\ddot{\theta} - b\sin\theta\,\dot{\theta}^2 = F, \tag{1}$$

$$\text{Pendulum (unactuated) coordinate:} \quad b\cos\theta\,\ddot{x} + c\,\ddot{\theta} - m_p\,g\,l\,\sin\theta = 0. \tag{2}$$

Energy Shaping and Damping Injection

The IDA-PBC approach assigns a desired closed-loop energy of the form

$$H_d(x, \theta, \dot{x}, \dot{\theta}) = \frac{1}{2} \begin{bmatrix} \dot{x} \\ \dot{\theta} \end{bmatrix}^T M(q) \begin{bmatrix} \dot{x} \\ \dot{\theta} \end{bmatrix} + V_d(x, \theta),$$

with a desired potential energy

$$V_d(x, \theta) = \frac{1}{2} k_x \, x^2 + \frac{1}{2} k_\theta \, \theta^2,$$

where the gains $k_x > 0$ and $k_\theta > 0$ are chosen so that the potential has a convex minimum at $(x, \theta) = (0, 0)$.

The target closed-loop dynamics are then designed as

$$M(q) \, \ddot{q} + K_d \, \dot{q} + \nabla V_d(q) = 0,$$

with a positive-definite damping matrix K_d.

Since only the x-coordinate is actuated, the matching condition is imposed on the x-dynamics. We first solve the unactuated (pendulum) equation (2) for $\ddot{\theta}$:

$$\ddot{\theta} = \frac{m_p \, g \, l \, \sin\theta - b \cos\theta \, \ddot{x}}{c}.$$

Substitute this expression into (1):

$$a \, \ddot{x} + b \cos\theta \left(\frac{m_p \, g \, l \, \sin\theta - b \cos\theta \, \ddot{x}}{c} \right) - b \sin\theta \, \dot{\theta}^2 = F.$$

Rearranging the terms, we obtain:

$$\left(a - \frac{b^2 \cos^2\theta}{c} \right) \ddot{x} + \frac{b \, m_p \, g \, l \, \cos\theta \, \sin\theta}{c} - b \sin\theta \, \dot{\theta}^2 = F.$$

If we assign the desired horizontal dynamics as a second-order linear oscillator

$$\ddot{x} + k_{dx} \, \dot{x} + k_x \, x = 0,$$

then

$$\ddot{x} = -k_{dx} \, \dot{x} - k_x \, x.$$

Substituting this into the previous expression yields the candidate control law:

$$F = \left(a - \frac{b^2 \cos^2\theta}{c} \right) \left(-k_{dx} \, \dot{x} - k_x \, x \right) + b \sin\theta \, \dot{\theta}^2 - \frac{b \, m_p \, g \, l \, \cos\theta \, \sin\theta}{c}.$$

Final IDA-PBC Controller

The IDA-PBC control law for the Segway-like system is thus given by

$$\boxed{F = -\left(a - \frac{b^2 \cos^2\theta}{c} \right) \left(k_{dx} \, \dot{x} + k_x \, x \right) + b \sin\theta \, \dot{\theta}^2 - \frac{b \, m_p \, g \, l \, \cos\theta \, \sin\theta}{c}.}$$

Explanation of the Terms

- **Decoupling Term:** The factor
$$\left(a - \frac{b^2 \cos^2\theta}{c}\right)$$
arises naturally when eliminating the unactuated coordinate θ and helps decouple the x-dynamics.

- **Desired Dynamics:** The term
$$-\left(a - \frac{b^2 \cos^2\theta}{c}\right)\left(k_{dx}\,\dot{x} + k_x\,x\right)$$
imposes the desired second-order linear behavior for x with gains k_{dx} and k_x.

- **Nonlinear Cancellation:** The terms
$$b\sin\theta\,\dot{\theta}^2 - \frac{b\,m_p\,g\,l\,\cos\theta\,\sin\theta}{c}$$
cancel the nonlinear coupling and gravitational effects arising in the dynamics.

Implementation and Tuning

1. **Tuning Gains:** Choose k_x and k_{dx} (and additional damping if necessary) so that the closed-loop poles lie in a desirable region. In simulation, start with moderate gains and adjust as needed.

2. **Validation:** Simulate the closed-loop system over a range of initial conditions to verify robustness, particularly because the design relies on cancellation of nonlinear terms in an underactuated system.

3. **Practical Considerations:** In real experiments, exact cancellation may not be feasible due to modeling errors. A robustifying term or approximate cancellation might be necessary.

Summary

The proposed IDA-PBC controller for the Segway-like system is:

$$\boxed{F = -\left(a - \frac{b^2 \cos^2\theta}{c}\right)\left(k_{dx}\,\dot{x} + k_x\,x\right) + b\sin\theta\,\dot{\theta}^2 - \frac{b\,m_p\,g\,l\,\cos\theta\,\sin\theta}{c}.}$$

This controller:

- Shapes the horizontal dynamics to behave as a linear oscillator.
- Cancels and shapes the coupling between x and θ.
- Injects damping through the k_{dx} term.

Implement and test this control law in your MATLAB simulation framework, and adjust the gains to achieve the desired transient performance and robustness.